Yuanbo Pang

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UC Berkeley EECS student building AI agent evaluation, infrastructure, and robotics/vision systems; co-first author on the NeurIPS 2026 Agent's Last Exam benchmark submission.

SKILLS

Languages: Python, C++, TypeScript/JavaScript, Java, Rust | **Robotics/Vision:** 3D perception, SLAM, UAV detection, optical flow, camera-motion compensation, OpenCV | **AI/ML:** agent harnesses, LLaVA, PyTorch, TensorFlow, Gemini API | **Infra:** FastAPI, React, Docker, Linux, Git, REST APIs, CI/CD, Firebase, Vercel, Modal

EDUCATION

University of California, Berkeley — **Berkeley, CA**
B.S. Electrical Engineering and Computer Sciences (EECS) — *Aug 2025 - Dec 2026*
- Coursework: CS 61B, CS 70, CS 170, EECS 16A, EECS 106A.

Stanford University, Summer Session — **Stanford, CA**
CS 229: Machine Learning — *Summer 2025*

Community College Coursework — **Cupertino, CA**
Computer Science coursework at De Anza, Foothill, Evergreen, and Diablo Valley; Phi Theta Kappa; Academic Director, Quantum Club; President, Data Club — *Sep 2023 - Jun 2025*

EXPERIENCE

Robotics AI Engineering Intern, Starbot — **Remote**
Robotics agent systems, task orchestration, service workflows — *Dec 2025 - Feb 2026*
- Built agent harness for a restaurant service robot, coordinating ordering, delivery, and customer-service tasks through structured robot workflows.
- Implemented tool interfaces, task-state logic, and recovery paths for reliable human-robot service interactions in demo conditions.
- System was demonstrated at CES 2026 and covered by NHK News, one of Japan's leading national media outlets.

Undergraduate Researcher, UC Berkeley - Prof. Avideh Zakhor — **Berkeley, CA**
UAV small-object detection, motion priors, airborne target tracking — *Mar 2026 - Present*
- Research small UAV detection and tracking in aerial video, focusing on tiny-object recognition, camera-motion compensation, and low-false-positive perception.
- Benchmarked UAV detection methods including TransVisDrone, AOT Winner v022, ESOD, and Li-TETC/NPS against 1,600 manually annotated DJI frames.
- Evaluated optical-flow and background-compensation pipelines to reduce false positives in tiny airborne object tracking.

Full-Stack Engineer, Ludus — **Remote**
AI agent marketplace, full-stack systems, agent execution — *Apr 2026 - Present*
- Built outcome-first marketplace where multiple coding/research agents execute the same task in parallel and users pay only for accepted results.
- Built React/TypeScript frontend, Node/Hono backend, Neon Postgres task store, and Docker-based VM dispatch layer for agent execution.

ENGR Team Co-Lead, UC Berkeley BAIR - Agent's Last Exam (ALE) — **Berkeley, CA**
Benchmark infrastructure, VM workflows, evaluation systems — *Jan 2026 - Present*
- Built infrastructure for reproducible agent benchmark runs across VM-based GUI/CLI workflows, task setup, artifact capture, and deliverable-based scoring.
- Converted expert-submitted tasks into executable benchmark instances spanning 55 sub-fields and 13 industry clusters.
- Co-first author on ALE, a NeurIPS 2026 benchmark submission for evaluating frontier AI agents on long-horizon professional workflows.

Software Engineering Intern, Geopogo — **Berkeley, CA**
AI rendering platform, full-stack production engineering — *Sep 2025 - Dec 2025*
- Shipped production AI architectural rendering platform as sole engineer, implementing React/TypeScript frontend, FastAPI backend, Firebase auth, and Vercel CI/CD.
- Integrated Gemini image reasoning and RunwayML text-to-video workflows with drag-and-drop uploads, real-time previews, and credit-based subscriptions.

Research Intern, Prof. Hao Wang - Stevens Institute of Technology — **Hoboken, NJ**
Personalized federated learning, differential privacy, PyTorch/TensorFlow — *Jun 2024 - Feb 2026*
- Built personalized federated learning systems with differential privacy guarantees in PyTorch/TensorFlow to protect local data while accelerating on-device model personalization.

PROJECTS

AI Mario Level Generator - mario-hackmit.vercel.app
HackMIT 2025 - Modal Sponsor Prize — *Sep 2025*
- Built sketch-to-playable-level pipeline using LLaVA 1.5, H100 GPU inference on Modal, FastAPI, React, and OpenCV.

Context8 - context8.org
Agent knowledge systems — *Oct 2025 - Jan 2026*
- Built self-evolving knowledge system for AI agents, focused on reusable debugging traces, structured feedback, and reliable agent workflows.

Stud.ai
HackMIT 2024 - Smartest AI Agent Award — *Sep 2024*
- Built AI planning agent that converts assignment rubrics into timelines and calendar work blocks.

PUBLICATIONS

Agent's Last Exam (ALE) - NeurIPS 2026 Evaluations and Datasets Track submission — **May 2026**
- Co-first author. Benchmark of 1,000+ real-world agent tasks across 55 sub-fields and 13 industry clusters, led by 300+ industry experts and Prof. Dawn Song, UC Berkeley.

AWARDS

Modal Sponsor Prize, HackMIT 2025; Smartest AI Agent Award, HackMIT 2024; Best Use of .Tech Domain, HackDavis 2024; Winner, Hack for Impact 2024; First Prize, Jiangsu Province Chinese National Physics Olympiad 2022.